

Build Update – June 21, 2026

On June 21, I completed the suspension installation and tested the first prototype of my custom RC crawler. During this initial test, only the front axle was connected because the rear driveshaft was too short to install.

While testing the vehicle in reverse, I quickly identified another issue: the front driveshaft was also too short. Under load, the suspension compressed and extended enough that the driveshaft disconnected from the transmission output.

Based on these observations, the next steps are:

- Install longer driveshafts to accommodate suspension travel.
- Adjust the shock preload to reduce chassis movement caused by drivetrain torque.
- Reevaluate driveshaft length and suspension geometry after the new components are installed.

To position the shocks accurately, I used a vernier caliper to measure and locate the mounting holes. Measurements were taken carefully on both sides to ensure the suspension remained as symmetrical as possible.

Key Lessons Learned

- Driveshaft length must account for the full range of suspension travel.
- Testing with only one driven axle can significantly affect vehicle dynamics.
- Precise measurement during fabrication is essential for proper suspension alignment and performance.